SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 2 – Concept Mapping (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO1	**Assessment:**	Assessment 2: Concept Mapping
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO1 Statement:	***Apply the fundamental concepts, image formation models, and image processing techniques to analyze and solve real-world Computer Vision problems.***		
Student Name:	Aasrika Thatamah'	**Reg. No.:**	192311008
Section / Batch:	CSE	**Date:**	9/1/2026

Code of Conduct

I 192311008 *(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Aasrika

QUESTIONS: 5

Total Marks: 50

Annexure A

Concept Mapping

Q1. An automobile company is developing an autonomous vehicle that uses cameras to detect vehicles, pedestrians, and traffic signs. Objects closer to the camera appear larger, while distant objects appear smaller. **Prepare a concept map applying the image formation and camera projection model to show how a three-dimensional road scene is converted into a two-dimensional image. Include the relationships between focal length, object distance, image distance, and projected image size.**

Q2. A manufacturing company uses cameras to measure the dimensions of bottles moving on a conveyor belt. Engineers observe that bottles at different distances from the camera appear to have different sizes. **Prepare a concept map applying perspective projection to show how focal length, camera position, viewing angle, object distance, and image size are related. Use the concept map to represent how these factors influence the accuracy of bottle measurement.**

Q3. A construction company uses a drone camera to capture images of buildings from different heights and viewing angles. The appearance and measured dimensions of the building change when the drone position is changed.
Prepare a concept map applying image formation and projection concepts to show the relationship between camera orientation, focal length, perspective projection, camera position, calibration, and the accuracy of building measurements.

Q4. An industrial company wants to select a suitable projection model for road monitoring and inspection of objects placed on a flat conveyor belt. The engineers are considering perspective and orthographic projection for these applications.
Prepare a concept map comparing perspective and orthographic projection based on depth, object size, distance, parallel lines, and image formation. Use the concept map to identify the appropriate projection model for each application.

Q5. A robotics company wants a robot to identify the position and shape of objects placed on a table using a camera. The camera captures only a two-dimensional representation of the three-dimensional objects. **Prepare a concept map applying the image formation model to show the relationship between a 3D object, camera parameters, projection geometry, focal length, camera orientation, and the resulting 2D image. Show how these concepts help determine the object's position and appearance.**

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Image formation and Camera projection in Autonomous Vehicles.

Image formation in Autonomous Vehicles

(3D road environment containing vehicles, pedestrians and traffic signs)

↓

1. Image Acquisition

(Cameras capture light reflected from real-world objects).

↓

2. Camera Parameters

[focal length (f). Object distance (z). Camera position. Orientation)

Focal length
(controls image magnification)

Object Distance.
(Greater distance generally means a smaller image).

↓

3. Perspective Projection

[(converts 3D coordinates (x, y, z) into 2D image coordinates (x, y)]

↓

4. Image formation and Preprocessing

(Pixel representation. Resizing. Noise reduction. Contrast enhancement

5. Object Detection and

(Identify vehicles, pedestrians, traffic signs and their loca

↓

6. Autonomous Driving Decisions.

(checks obstacle avoidance. Pedestrian safety. Traffic sign recognitio

Mathematical relationship :-

$$x' = \frac{fx}{z}, \quad y' = \frac{fy}{z}.$$

Perspective Projection for Bottle Management.

Bottle Measurement System.

(Bottles moving along a conveyor belt).

↓

1. Camera setup

(Focal length. Camera height. Position. Viewing angle).

Focal length
(changes projected image scale)

Viewing Angle
(May distort apparent height and width).

↓

2. Perpective Projection

(Maps real bottle dimensions into the 2D image plane).

3. Image Measurement

(Detect bottle boundaries and measure pixel height and width).

↓

4. Camera Calibration.

Determine cameras parameters and pixel -

↓

5. Real world Dimension Estimation

(Calculate actual height and width using calibrated geometry).

↓

6. Quality Inspection

(Detect incorrect dimensions and defective bottles)

Projected height :-

$$hi = \frac{fho}{Z}$$

Actual height : ho , projected height, hi ; distance : Z.

Factors affecting accuracy :-

↳ Different bottle distances change apparent image size.

↳ Camera tilt affects the projected dimensions.

↳ Incorrect calibration produces measurement errors.

↳ Blur and poor lighting make bottle boundaries difficult to detect.

3D Building

(Actual dimensions, surfaces and geometric structure)

↓

1. Drone Image Acquisition

(capture building images at different heights and viewing angles).

↓

2. Camera Parameters

focal length. camera position. Flight height. Orientation.

Camera Position

(change camera-to-building distance)

Camera Orientation.

(changes perspective and visible surfaces).

↓

3. Perspective Projection

(converts 3D building coordinates into a 2D image).

↓

4. Image Processing and Calliboration

(Feature detection. Lense distortion correction. Camera callibration)

↓

5. Dimension Estimation

(estimate building height

6. Building Inspection and 3D reconstruction.

Projection equation :-

$$\left[hi = \frac{fho}{z} \right]$$

* Factors affecting accuracy :-

↳ changing flight height changes object distance and projected size.

↳ Camera rotation and tilt alter the building's appearance.

↳ Lens distoration may cause errors in measured dimensions.

↳ Multiple calibrated images help reconstruct building geometry.

Perspective Projection vs orthographic projection :-

& 3D Scene

(objects with depth, size and spatial position)

↓

Projection Model selection

Perspective Projection

Finite camera centre

Depth affects projected size

Distant objects appear smaller

Parallel lines may change

Realistic depth appearance

orthographic Projection

Parallel projection rays

Depth does not affect projected.

Parallel lines remain silent

Reduced perpective distoration

Useful for technical measurements.

Equation

$$x' = \frac{fx}{z}$$

$$y' = \frac{fy}{z}$$

Applications

- Road monitoring
- Automous vehicles

Equation

$$x' = sx$$

$$y' = sy$$

Applications

- Engineering drawings
- Flat conveyor inspectio

↓

Comparision and Selection :-

(Perpective → Realistic scenes with varying depth → orthographic → Consistent scale for suitable measurement views)

Image formation for Robotic object Identification :-

3D object on a Table.

(Shape . Dimensions . Position . Orientation)

↓

1. Camera Parameters

(focal length . Principal point . Camera position)

Intrinsic Parameters :-

scribe internal Camera characteristics)

Entrinsic Parameters :-

(Rotation (R) and

2. Projection Geometry

(Maps 3D coordinates to 2D image coordinates)

↓

3. Image formations

(Pixel values . Edges . Colour . Texture . Projected shape)

↓

4. Computer Vision Processing

(Object detection . Feature extraction . Shape recognition)

↓

5. Position and orientation Estimation

(Calibration . Table geometry . Depth information)

↓

6. Robotic Action

(Object grasping . Pick and place . Collision avoidance)

Camera projection model :-

$$sp = K [R \; t] P.$$

K = intrinsic matrix

R = Rotation

t = transition

P = 3D point.